

ABHIJEET GUPTA

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Quantitative Research | Financial Modeling | Time Series | Statistical Backtesting

SUMMARY

CS PhD candidate graduating May 2027 building reproducible quantitative research pipelines for derivatives pricing, volatility forecasting, market-neutral backtesting, and machine-learning forecasting. Each public repository includes frozen data or manifests, leakage tests, baselines, confidence intervals, machine-readable results, and local M3 Max reproduction evidence.

TECHNICAL SKILLS

Quantitative: probability, statistics, stochastic processes, Black-Scholes, Monte Carlo, Greeks, implied volatility, GARCH(1,1), EWMA, Engle-Granger cointegration, bootstrap inference

Engineering: Python, PyTorch, NumPy, pandas, SciPy, statsmodels, scikit-learn, yfinance, walk-forward validation, leakage testing, GitHub Actions, uv

VERIFIED QUANTITATIVE PROJECTS

Options Pricing and Greeks | June 2026 reproduction

- Implemented Black-Scholes calls/puts, variance-reduced Monte Carlo, analytical and finite-difference Delta/Gamma/Vega/Theta/Rho, implied volatility, and arbitrage properties.
- At the benchmark configuration, largest Monte Carlo pricing error was 0.00866 versus 0.00893 standard error; maximum finite-difference Greek error was $6.15e-9$.

Statistical Pairs Trading | June 2026 reproduction

- Built lagged-signal walk-forward research for AAPL-MSFT, KO-PEP, and XOM-CVX with frozen daily snapshots, rolling hedge ratios, costs, slippage, bootstrap Sharpe intervals, and drawdown.
- KO-PEP was the only pair cointegrated at 5% ($p=0.029$); its Sharpe was 0.212 with a 95% bootstrap interval of $[-0.465, 0.890]$, explicitly showing high uncertainty.

GARCH Volatility Modeling | June 2026 reproduction

- Compared one-day GARCH(1,1), EWMA, and rolling-volatility forecasts on SPY and five equities using MAE, RMSE, QLIKE, residual diagnostics, and strictly historical refits.
- GARCH had lowest RMSE for JNJ, JPM, SPY, and XOM; rolling volatility led AAPL and MSFT. The published conclusion is model-dependent rather than a universal GARCH win.

Financial Time-Series Forecasting | June 2026 reproduction

- Implemented chronological PyTorch LSTM evaluation against random-walk, seasonal-naive, ridge, and random-forest baselines with train-only scaling and alignment tests.
- LSTM achieved the best RMSE (1.186) and MAE (0.880), but directional accuracy was 0.492, below random walk (0.506) and seasonal naive (0.512); no trading edge is claimed.

EXPERIENCE

Graduate Research Assistant | University of Dayton | Aug 2023 - Present

- Builds controlled ML evaluation pipelines spanning adversarial reinforcement learning, computer vision, and reasoning-model analysis; coordinates reproducible Slurm and multi-GPU experiments.
- Co-authored peer-reviewed ML/CV research and mentors junior researchers on experiment design, PyTorch implementation, statistical validation, and publication artifacts.

Data & Systems Engineering Intern | Dayton Phoenix Group | Jan 2025 - Present

- Built Python data pipelines and KPI reporting for manufacturing systems; applied statistical process control and anomaly analysis to production metrics.

Software Engineer | Walmart Global Tech via Kitestring | Apr 2023 - Aug 2023

- Built Docker/Kubernetes CI/CD workflows for a Node.js point-of-sale application and Python validation tooling for migration of more than one million records to Azure Cosmos DB.

ML / Product Intern | Walmart | Jun 2022 - Aug 2022

- Developed a Random Forest supply-chain classifier and integrated heterogeneous operational data sources with leakage-aware validation.

EDUCATION

PhD, Computer Science | University of Dayton | Expected May 2027

Research: LLM evaluation and interpretability, reinforcement learning, computer vision, and reproducible machine learning.

MS, Computer Science | University of Texas at Arlington | 2022

BE, Computer Science and Engineering | RTM Nagpur University | 2019

RESEARCH STANDARDS

- Repositories support uv sync, make test, make reproduce-smoke, and make reproduce-results.
- Results are generated from committed configurations and include environment records, data manifests, predictions, statistical tests, figures, and explicit negative findings.
- Public claims are limited to evidence reproduced on June 11, 2026; unavailable experiments are marked NOT_RUN rather than inferred or invented.